

AI Customer Support Agent for @AppleSupport

Hiver SDE Intern Take-Home Technical Report | Evaluation-Focused AI Prototype

Executive Summary

Social media customer support on Twitter/X presents acute operational constraints: interactions are public, real-time, and capped at 280 characters. For a premier consumer brand like Apple, an AI agent must not only provide high-accuracy technical guidance; it must know with statistical reliability **when not to answer**. This report presents an evaluation-focused AI Support Agent prototype for @AppleSupport, benchmarked on a 200-sample hand-audited, 100% genuine holdout Golden Evaluation Set. The holdout gold set is strictly thread-disjoint from both the 600 training threads and the 1,000 historical KB retrieval corpus (with training threads contained within the KB). Across this benchmark, the proposed agent achieves: **62.0% Out-of-Sample Intent Accuracy** across a 7-class domain taxonomy (vs. 35.5% Trivial, 55.0% Simple Baseline), **69.2% Escalation Recall** on safety-critical interactions (vs. **0.0%** for Trivial and **3.9%** for Simple Baseline), **79.5% Official Apple Domain Link Validity**, and **100% Twitter character compliance**. The full evaluation harness reproduces deterministically on standard CPU in **6.3 seconds**.

1. Problem Framing & Scope Boundaries

Operational Principles for @AppleSupport:

- **Safety and Privacy First:** Public social threads must never handle account credentials, passwords, 2FA codes, or billing card details. The agent immediately escalates to secure DM routing and authenticated recovery portals.
- **Definitive Actionability:** Concrete, verifiable UI paths (e.g. *Settings > Battery > Battery Health*) accompanied by verified canonical Apple URLs (support.apple.com/HT...).
- **Apple Brand Persona:** Calm, professional, empathetic tone within Twitter's 280-character limit.

Explicit Non-Goals:

- *No Automated Financial Payouts:* All refund and subscription cancellation requests require authenticated human review to prevent social media bot fraud.
- *No In-Chat Credential Resets:* Account lockouts route exclusively to iforgot.apple.com.
- *No Machine-Translated Technical Troubleshooting:* Non-English queries are routed to native-language queues to prevent dangerous mistranslation of recovery procedures.
- *No Speculative Hardware Cost Estimation:* Physical component failures route to official Genius Bar scheduling.

2. Empirical Results vs. Two Baselines

Comparative performance on N=200 Golden Evaluation Set (Thread-Disjoint Holdout Split):

Metric Dimension	Trivial Baseline (Always Auto-Handle)	Simple Baseline (Naive Bayes + 1-NN)	Proposed AI Agent (RAG + Policy Engine)
Intent Accuracy (Out-of-Sample)	35.5%	55.0%	62.0%
Intent Macro F1	0.075	0.287	0.476
Brier Calibration Score (Lower=Better)	1.0	0.7919	0.6049
Expected Calibration Error (ECE)	0.355	0.3573	0.0453
Escalation Accuracy	74.0%	75.0%	71.0%
Escalation Recall (Safety-Critical)	0.0%	3.9%	69.2%
Escalation Precision	0.0%	100.0%	46.2%
Escalation F2 Score (Recall-Weighted)	0.000	0.048	0.629
False Escalation Rate (Lower=Better)	0.0%	0.0%	28.4%
Illustrative 5:1 Risk Penalty (5*FN + 1*FP)	260	250	122
SacreBLEU Score	0.2	0.4	4.1
Twitter Char Limit Compliance (<280)	100.0%	95.0%	100.0%
Official Domain Inclusion Rate	0.0%	0.0%	79.5%
Intent-Link Relevance Rate	0.0%	0.0%	75.8%

Heuristic: Groundedness (1-5)	4.20	4.06	4.78
Heuristic: Brand Voice & Empathy (1-5)	5.00	4.36	4.58
Heuristic: Actionability (1-5)	4.20	4.33	4.80
Heuristic: Escalation Appropriateness	4.14	4.18	4.42
Heuristic: Overall Quality Score (1-5)	4.39	4.24	4.65

The Illusion of Baseline Accuracy: The Trivial Baseline achieves 74.0% accuracy simply by auto-handling all inquiries. However, its **Escalation Recall is 0.0%**, missing every safety-critical ticket. The Proposed Agent catches **69.2% of escalations** (36 of 52), reducing the illustrative 5:1 weighted risk penalty from 260 to 122.

3. Failure Mode Analysis (Top 5 Real Observed Failures)

Failure Mode 1: Sarcasm and Ambiguous Frustration [GOLD_002]

- *Query:* "It's been nearly two weeks and I've yet to get LTE on my Watch () working. Think I paid a premium for a spec of red paint."
- *Observed Output:* Predicted Intent = OUT_OF_SCOPE_OTHER (Conf: 0.49). Decision = AUTO_HANDLE.
- *Gold:* Intent = CUSTOMER_FEEDBACK_COMPLAINT, Escalation = ESCALATE.
- *Root Cause & Mitigation:* Sarcastic complaint lacked explicit vulgarity or legal threats, and confidence (0.49) was above the low-confidence threshold (0.40). Mitigation: Integrate multi-week duration and churn phrase detectors.

Failure Mode 2: Multilingual Language Routing Miss [GOLD_004]

- *Query:* "Mira que me gusta vuestra actualización pero me va como el culo ahora, cuando queráis lo solucionáis."
- *Observed Output:* Predicted Intent = OUT_OF_SCOPE_OTHER (Conf: 0.69), Decision = AUTO_HANDLE.
- *Gold:* Intent = OUT_OF_SCOPE_OTHER, Escalation = ESCALATE (Non-English routing).
- *Root Cause & Mitigation:* Idiomatic Spanish ('como el culo') evaded the localization regex, auto-handling in English. Mitigation: Pre-classification language identification filter (fastText/langdetect).

Failure Mode 3: False Escalation on Repeated Troubleshooting Phrasing [GOLD_008]

- *Query:* "I've had to do it multiple times when I reset my device and how do I change trusted device from old phone to new phone?"
- *Observed Output:* Decision = ESCALATE (via POLICY_REPEATED_UNRESOLVED_FAILURE).
- *Gold:* Decision = AUTO_HANDLE (Routine device pairing how-to).
- *Root Cause & Mitigation:* The phrase 'multiple times' triggered the repeated failure heuristic on a standard UI inquiry. Mitigation: Condition repeated-failure rules on negative emotional sentiment tokens.

Failure Mode 4: Low-Confidence Safety Interception [GOLD_019]

- *Query:* "Cool new feature in macOS High Sierra, it knows you've been working too hard and freezes the screen, but not the mouse."
- *Observed Output:* Intent = IOS_SOFTWARE_UPDATE (Conf: 0.37). Escalation = ESCALATE (via POLICY_LOW_MODEL_CONFIDENCE).
- *Gold:* Intent = IOS_SOFTWARE_UPDATE, Decision = AUTO_HANDLE.
- *Root Cause & Mitigation:* Low classifier confidence (< 0.40) safely prevented automated troubleshooting on a sarcastic OS freeze complaint, routing to human support.

Failure Mode 5: Sensitive Domain Precautionary Escalation [GOLD_022]

- *Query:* "Grrr! Why won't the contacts from my phone back up to iCloud?? #icloud"
- *Observed Output:* Intent = APPLE_ID_AND_ICLOUD (Conf: 0.77). Escalation = ESCALATE (via POLICY_SENSITIVE_DOMAIN).
- *Gold:* Intent = APPLE_ID_AND_ICLOUD, Escalation = AUTO_HANDLE.
- *Root Cause & Mitigation:* Broad sensitive domain rules escalate all Apple ID/iCloud inquiries to safeguard account privacy, causing false escalation on routine contact syncing.

4. 'What is Misleading About My Headline Number?'

A responsible engineering evaluation must transparently acknowledge the limitations of offline benchmark figures:

1. **Single-Turn Static vs. Multi-Turn Dynamic Evaluation:** Our benchmark evaluates the incoming customer message in isolation. In production, customer interactions span multiple back-and-forth turns.
2. **Illustrative Asymmetric Cost:** Under an illustrative 5:1 penalty weighting missed escalations more heavily than unnecessary escalations, the policy reduces the penalty from 260 to 122. However, this is a candidate-selected metric, not an empirical dollar figure.
3. **Domain Taxonomy Conditioning:** Achieving 62.0% out-of-sample intent accuracy reflects a closed 7-class taxonomy. In an unconstrained open-vocabulary setting, intent accuracy would degrade.
4. **69.2% Escalation Recall in Production:** Catching 36 of 52 escalations means 16 are missed. In production, we would not deploy unrestricted auto-reply at this threshold; initial rollout must operate as an agent-assist copilot.

5. Candidate Human Annotator vs. LLM-as-a-Judge Agreement (N=50)

To assess automated grading reliability, we conducted an inter-rater agreement study comparing Gemini 2.5 Flash rubric scoring and candidate author blind scoring on the exact same 50 frozen agent outputs: **Pearson Correlation $r = 0.920$, Spearman $\rho = 0.766$, Mean Absolute Error = 0.226 points** (on 1–5 scale), with **100.0% of ratings within 0.5 points**. Cohen's κ is reported as N/A on categories with uniform scores ($\kappa = 1.000$ on binary escalation). Groundedness achieves 100% agreement within 0.5 points (MAE = 0.360) on official Apple URLs. SHA256 validation ensures both rating files refer strictly to identical frozen outputs.

6. What We Would Do Next with One More Week

1. **Tri-State Copilot Routing:** High confidence (>0.85) -> automated reply; Medium confidence ($0.55\text{--}0.85$) -> draft in Hiver inbox for one-click human agent approval; Low confidence (<0.55) -> direct human routing.
2. **Stateful Multi-Turn Conversation Memory:** Track user dialogue history across multiple tweets to disambiguate intermittent hardware vs. software symptoms.
3. **Mock CRM Tool Calling:** Query Apple System Status API and serial-number warranty entitlement endpoints prior to drafting responses.